26/10/26 – TRAFFIC CONTROL

$$J = \underbrace{\sum_{i=0}^{W-1} \sum_{j=1}^{2} \left(n_{ik,j}\right)^2}_{J^m \text{ MIN. AND BALANCE QUEUES}} + \underbrace{\qquad}_{J_W}$$

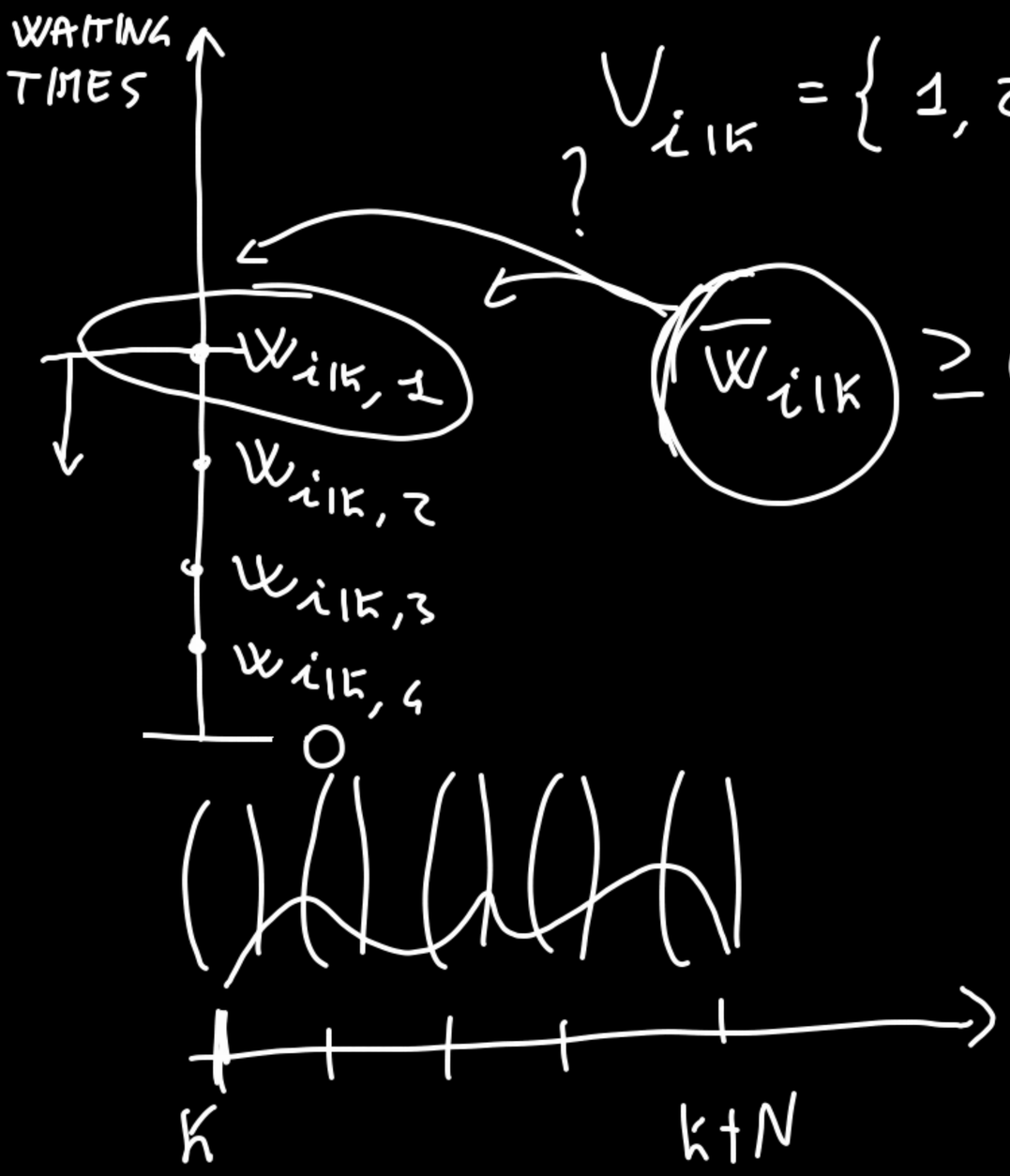
WAITING TIMES: MINIMIZE THE MAXIMUM WAITING TIME.

$W_{ik,v}$, V_{ik} SET OF VEHICLES PRESENT AT TIME ik.

$V_{ik,j}$

$W_{ik,j} \geq 0$

MINIMIZE THE MAXIMUM WAITING TIME

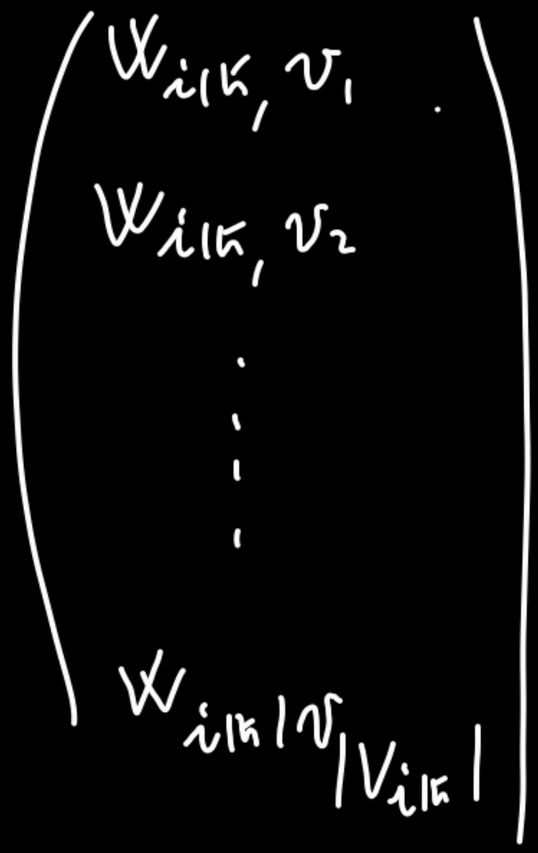


$V_{ilk} = \{1, 2, 3, 4\}$

?

$\overline{W}_{ilk} \geq 0$.

$ilk \longrightarrow$

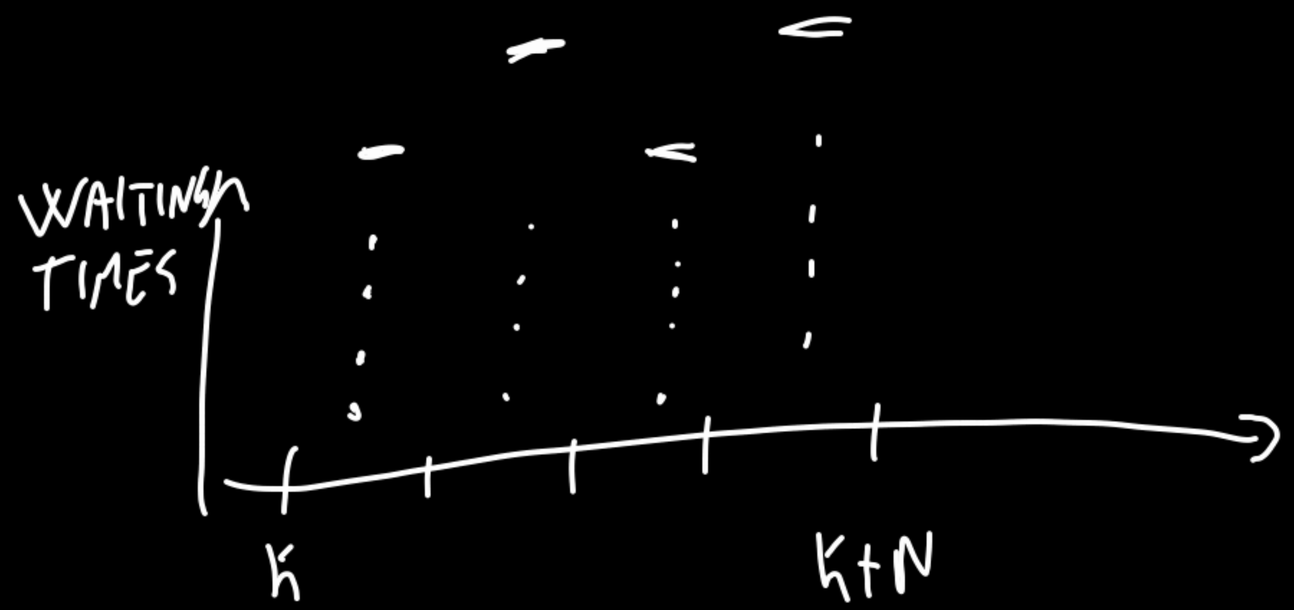


$$\|v\|_\infty = \max_J |V_J|$$

$$\overline{W}_{i|k} \geq 0 \quad , \quad i = 0, 1, \dots, N-1$$

$$W_{i|k,v} \leq \overline{W}_{i|k} \quad , \quad i = 0, 1, \dots, N-1 \quad , \quad \forall v \in \mathcal{V}_{i|k}$$

$\overline{W}_{i|k}$ IS AN UPPER BOUND OF ALL WAITING TIMES



$$J^{W} = \sum_{i=0}^{N-1} \left(\overline{W}_{i|k} \right)^2$$

CONSTRAINT ON YELLOW SIGN DURATION

$$\left(\text{IF } y_{i1k,J} = 1\right) \wedge \left(t^{y}_{i1k,J} < t^{min}_{J}\right) \longrightarrow \left(r_{i+1|k,J} = 0\right)$$

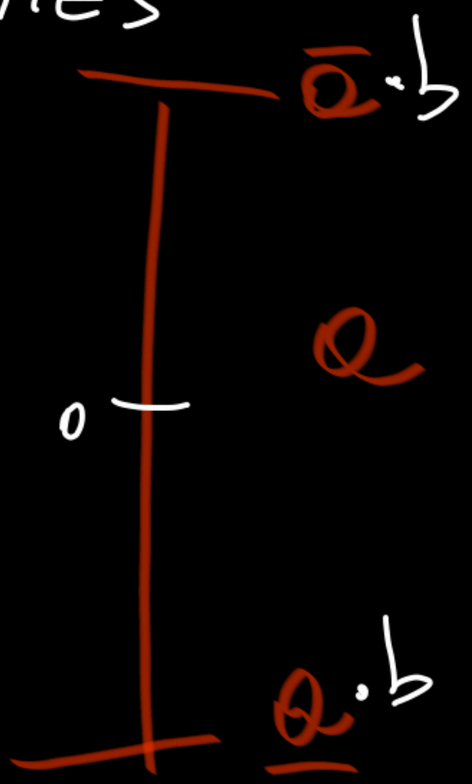
FORMULA THAT [0,1) WHEN CONDTION IS TRUE

$$r_{i+1|k,J} \leq \left(\frac{t^{y}_{i1k,J}}{t^{min}_{J}}\right) + \left(1 - y_{i1k,J}\right), \forall i,J$$

when $y_{i1k,J} = 0$, CONSTRAINT IS IRRELEVANT.

LINEARIZATION OF PRODUCT OF REAL VARIABLE TIMES BOOLEAN VARIABLE.

$$t^{y}_{i+1|k,j} = \left(t^{y}_{i|k,j} + T\right) \cdot y_{i|k,j} \quad , \forall i,j$$



$a \in R$, $b \in \{0,1\}$. HOW TO LINEARIZE $a \cdot b$.

$p \in R$ p SHOULD BE EQUAL TO $\underline{a} \cdot b$.

② WRITE A CONSTRAINT THAT WHEN $b=0$ WILL FORCE p TO ZERO,

$$\underline{a} \cdot b \leq \underline{p} \leq \bar{a} \cdot b$$

(2) WRITE A CONSTRAINT THAT WILL FORCE $p=q$ WHEN $b=1$.

$q-1 \leq p \leq q+1$

$$q - \underline{M}\underline{q}\underline{(1-b)} \leq p \leq q + \underline{M}\bar{q}\left(\underline{1-b}\right)$$

$(-3 \leq p \leq 3)$

WRITE A FORMULA THAT IS EQUAL TO q WHEN $b=1$ (AND SUCH THAT WHOLE CONSTRAINT IS IRRELEVANT WHEN $b=0$).

ROME

COUPLING CONSTRAINTS

$if\left(m_{ilk,T}\ ;\ W_{ilk,v}\ ;\ \begin{pmatrix} g_{ilk,T} \\ y_{ilk,T} \\ r_{ilk,T} \end{pmatrix}^{*}\ ;\ t^{y}_{ilk,T}\ ;\ \begin{pmatrix} p_{ilk,v} \\ v_{ilk,v} \end{pmatrix}\right.$

MPC

$d_1\ d_2$

g^{*}
y^{*}
r^{*}

DCDC

LDD

LAST EXAMPLE ASSEMBLY LINE CONTROL.

JOB: TO BUILD A VEHICCE IN AN ASSEMBLY LINE.

TASKS: ELEMENTARY OPERATIONS THAT MAKE THE JOB.

WORKSTATIONS: LOCATION OR MACHINE WHERE TASKS ARE EXECUTED.

RESOURCES: TO EXECUTE, TASKS NEEDS RESOURCES.

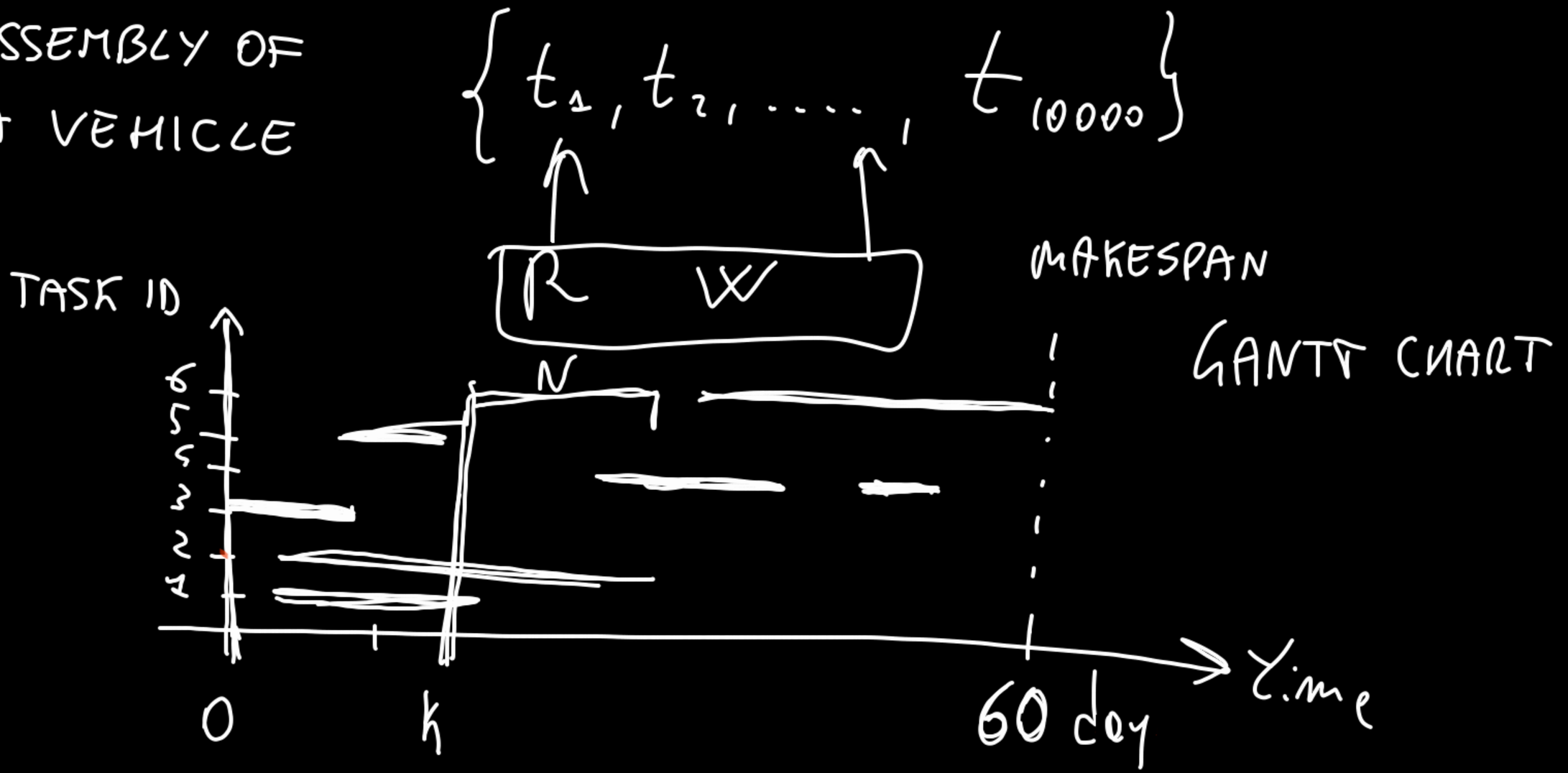
POSSIBLE CONTROL GOALS:

- MINIMIZE COST OF ASSEMBLY;
- MAXIMIZE QUALITY OF END PRODUCT;] DETAIL MODELS
- MINIMIZE SCRAP;
- MINIMIZE TOTAL ASSEMBLY TIME (MAKESPAN)
- BALANCE RESOURCE / WORKSTATION USAGE.

CONTROL GOAL: DECIDE A SCHEDULING OF THE TASKS TO MINIMIZE TOTAL COMPLETION TIME

ASSEMBLY OF A VEHICLE

$\{t_1, t_2, \ldots, t_{10000}\}$



MATHEMATICAL MODELING

AT ANY TIME, MPC SHOULD DECIDE WHICH TASKS TO EXECUTE, ON WHICH WORKSTATIONS AND USING WHICH RESOURCES.

$\mathcal{T}$ SET OF ALL TASKS

W SET OF ALL WORKSTATIONS

$\forall t,\ W_t \subseteq W$ SET OF WORKSTATIONS THAT CAN EXECUTE t

EX: $W_2 = \{3, 10, 51\}$

CONTROL VARIABLE FOR TASK EXECUTION



VARIABLE TO MODEL DECISION TO RUN (OR NOT) A GIVEN TASK, AT A GIVEN TIME, ON A GIVEN WORKSTATION.

$c_{i|k,t,w} \in \{0,1\}$

- $1 \Leftrightarrow$ AT TIME $i|k$, EXECUTE TASK t ON WORKSTATION $w \in \mathcal{W}_t$
- $0 \Leftrightarrow$ DON'T RUN TASK t AT $i|k$.

EXAMPLE: $c_{21K,3,10} = 1$

TASK PARAMETERS:

$\forall t$: ② S_t EARLIEST POSSIBLE START TIME.

② F_t DEADLINE.



③ d_t ESTIMATE FOR TOTAL TIME TO COMPLETE TASK. $(d_{10} = 3.5\,h)$

④ W_t SET OF WORKSTATIONS THAT CAN EXECUTE t

(5) (RESOURCES / TOOLS NEEDED TO EXECUTE TASK).

R SET OF RESOURCE TYPES

$R^{\#}_{t,r}$ NUMBER (#) OF RESOURCES OF TYPE r NEEDED TO EXECUTE TASK t.

$R^{\#}_{2,r} = (0, 1, 0, 3, 0, \ldots, 1)$

- 0 → WELDING MACHINE
- 1 → TOOL 1

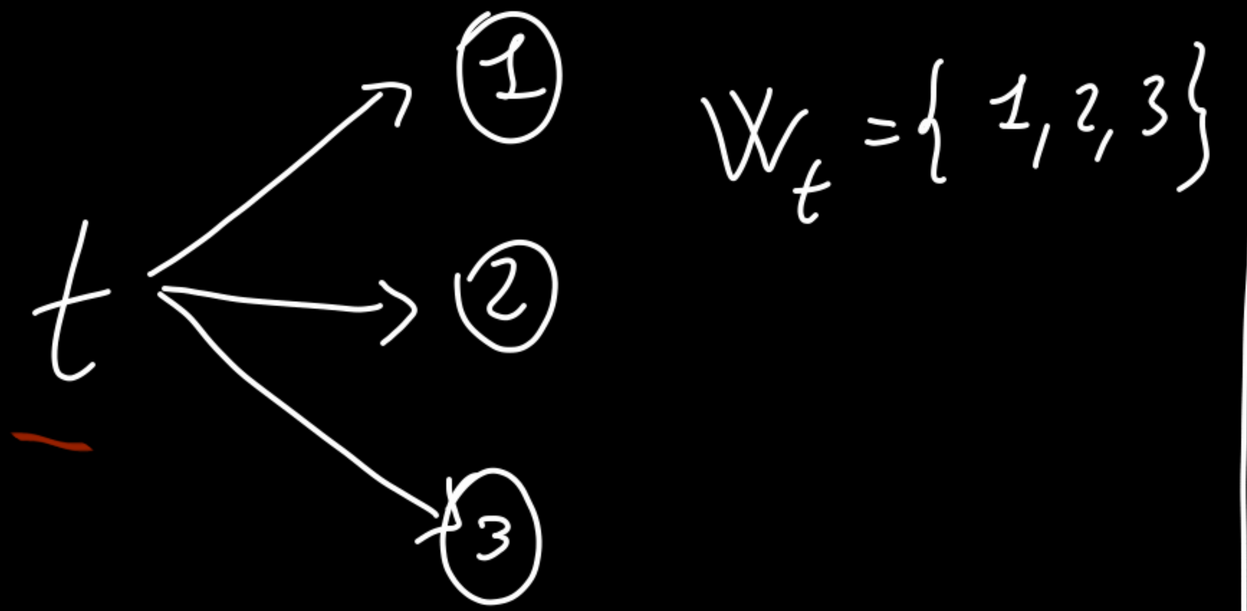
$R^{\#}_{r} = (100, 10, \ldots\ldots)$

TASK 2 NEEDS 0 ITEMS OF RESOURCE TYPE 1, 1 ITEM OF RESOURCE 2,

TASK CONSTRAINTS:

(MAIN VARIABLE

$c_{ilk,t,w} = 1$ RUN t AT ilk ON w)



ilk

CONSTRAINT 1: AT ANY TIME, A TASK CAN RUN ONLY ON ONE WORKSTATION.

$$\sum_{w \in W_t} c_{ilk,t,w} \leq 1, \quad \forall i,t$$

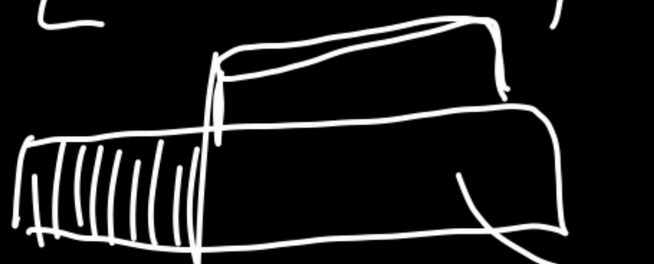
AUXILIARY VARIABLE

$$c_{ilk,t} = \sum_{w \in W_t} c_{ilk,t,w} \quad , \forall i,t$$

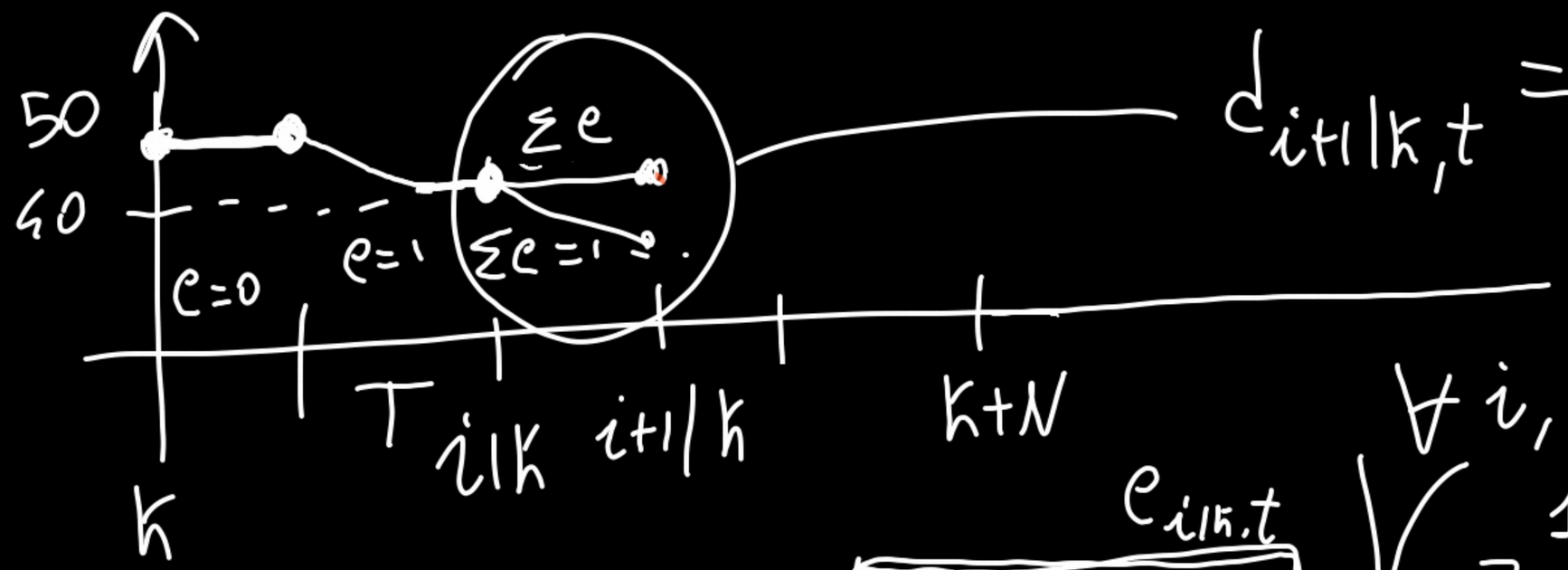
$c_{ilk,t} \in \{0,1\}$ — 1 IF t IS RUN AT ilk; 0 OTHERWISE

MODELING OF THE EVOLUTION OF TIME LEFT TO COMPLETE TASK.

d_t [TIME STEPS], EX: $d_2 = 51$ WE NEED TO WORK 51 TIME INTERVALS TO COMPLETE TASK ?.



$d_{ilk,t}$ PREDICTED TIME LEFT TO COMPLETE t AT ilk (STATE VARIABLE)



$$d_{i+1|k,t} = d_{ilk,t} - \underbrace{\sum_{w \in W_t} c_{ilk,t,w}}_{c_{ilk,t}} \quad \forall i,t$$

$c_{ilk,t}$: 1 IF TASK IS RUN AT ilk; 0 OTHERWISE

Workst

$t = 2$

→ (20) $c_{ilk,2,20} = 0$

→ (13) $c_{ilk,2,13} = 1$

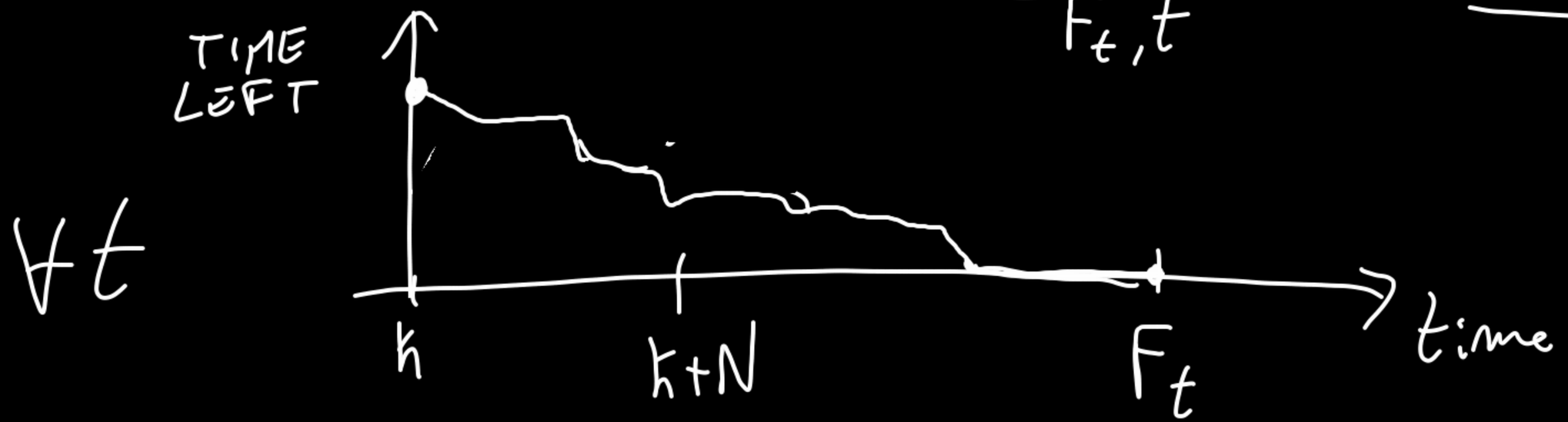
→ (3) $c_{ilk,2,3} = 0$

ilk

CONSTRAINT ON TASK DEADLINE.

F_t TASK DEADLINE (t MUST BE COMPLETED BEFORE F_t)

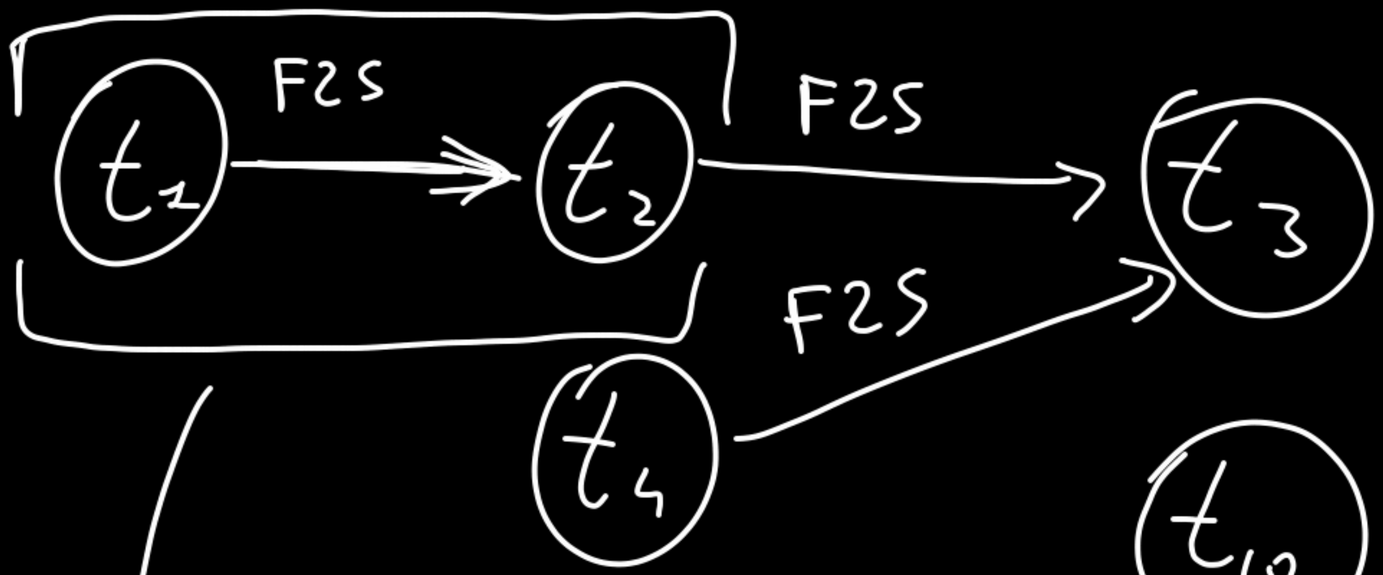
$$d_{F_t,t} = 0 \;,\; \forall t$$



$$d_{F_t,t} = 0 \;,\; \forall t : F_t \leq t+N.$$

PRECEDENCE CONSTRAINTS:

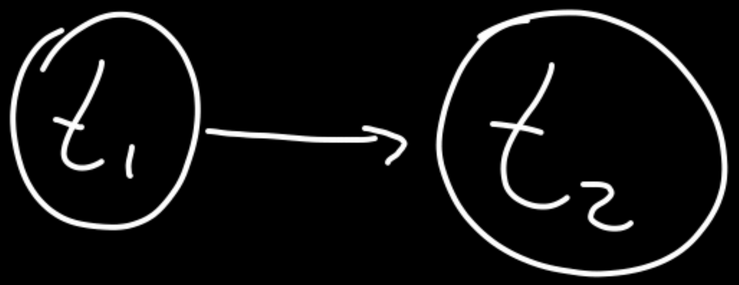
- F2S (FINISH TO START CONSTRAINT)



- F2F
- S2S
- S2F

→ WE MUST FINSH t_1 BEFORE WE CAN START t_2.

F2S



- WE MUST FINISH t_1 BEFORE WE CAN START t_2
- WE CANNOT (RUN) START t_2 UNTIL t_1 IS FINISHED.

$$\text{———} \leq \text{———}$$